

f-string
syntax: `f" { } { } "`
 ↑ ↑
 code/variable

format decimal place: `{n:.2f}`
 2 decimal

Ex `print(f"Total is {10:.2f}")`

↳ Total is 10.00.

$n=10$
`print(f" {10+13} and {2+3} or {n*100} ")`
 ↳ 11 and 5 or 100

Named Constants

↳ Variables in UPPERCASE that hold a constant value.

e.g. `RATE = 10`

Decision Structures

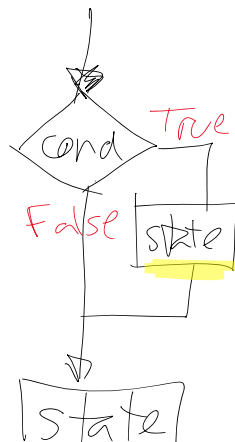
Comparison Operators

`>` is ~~x~~ greater `<`
`>=`
`<=`
~~`==`~~ equal
~~`!=`~~ not equal!
 outcome will be either True or False

If structure

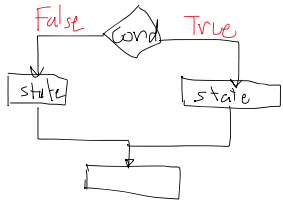
2 tab
 2 spaces
 if condition:
 statement
 statement

Ex `if 5 > 2:`
 `print('Hey')`
 `print('Done')`
 ↳ Hey
 ↳ Done



`print('Done')` → Done ↓
state

If-else Structure



if condition: True
 statement if-block
 statement
else: False
 statement else-block
 statement

* Logic: if condition is True, do if-block
 else (condition is False), do else-block.

Ex `x = 10`
 if `x > 5`: 10 > 5 → True
 `print('A')`
 else:
 `print('B')`
 `print('Done')`
Output: A
 Done

`x = 10`
 if `x == 5`: 10 == 5 → False
 `print('A')`
 else:
 `print('B')`
 `print('Done')`
Output: B
 Done

Donut Shop

- sell donut 20 baht each
- ask user for #donut
- get a discount if you buy at least 200 baht (-10)
- Need to print out before and after dis (2 dec)

`D_P = 20`

`d_n = int(input("Enter #donut:"))`

`total = d_n * D_P`

if `total >= 200`:

`discount = 10`

else:

`discount = 0`

`print(f"{total:.2f}")`

`print(f"{total-discount:.2f}")`

discount = 0

if `total >= 200`:

`discount = 10`

Nested Conditions

Loan Qualification

- The program approves or rejects the loan app.
- Approve if both conditions are True:
 - Salary at least 20000 baht = cond1
 - Been employed at least 2 years = cond2

salary = float(input(...)) = 30000 if cond1 and cond2
employed = int(input(...)) = 5

```
if salary >= 20000: → 30000 >= 20000 → T
    if employed >= 2: ← 5 >= 2 → T
        print("Approve")
    else:
        print("Reject")
else:
    print("Reject")
```

Logical Operators (Logic)

↳ operate True, false
↳ return True, false

$\wedge, \vee, \rightarrow, \neg, \leftrightarrow$

and operator ($\wedge$)

X	Y	X and Y
T	T	T
T	F	F
F	T	F
F	F	F

or operator ($\vee$)

X	Y	X or Y
T	T	T
T	F	T
F	T	T
F	F	F

not

X	not X
T	F
F	T

EX $x = \text{True or False} \rightarrow \text{True}$
 $y = \text{True and False} \rightarrow \text{False}$
 $\text{print(f" {x} {y} {not x} ")}$
↳ True False False

EX $a = (10 < 1) > (12 * 2) \text{ or } (10 + 2) == 5 \rightarrow \text{False}$
 $b = (a \text{ or True}) \text{ or } (\text{False and } (10 > 5))$
if $a \text{ or } b$: True
print('A')
else:
print('B')


```

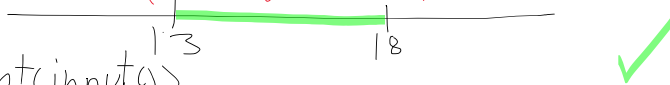
Salary = float(input())
employee = int(input())
cond1 = salary >= 20000
cond2 = employee >= 2
if cond1 and cond2: → (salary >= 20000) and (employee >= 2)
    print("Approve")
else:
    print("Reject")

```

Check interval

Is x between a and b ?

e.g. is age between 13 and 18?

$(\geq 13 \text{ and } \leq 18) \rightarrow$

 age = int(input())
 if age ≥ 13 and age ≤ 18 : → $13 \leq \text{age} \leq 18$

Grading Question

• Map score to a letter grade

- 'A' if score ≥ 80
- 'B' if $80 > \text{score} \geq 70$
- 'C' if $70 > \text{score} \geq 60$
- 'D' if score < 60

Only map to one letter

```

score = int(input())
✓ if score >= 80:
    print('A')
✓ if 80 > score >= 70:
    print('B')
✓ if 70 >= score >= 60:
    print('C')
✓ if score < 60:
    print('D')

```

```

if score >= 80:
    print('A')
else:
    if score >= 70:
        print('B')
    else:
        if score >= 60:
            print('C')
        else:
            print('D')

```

$\text{score} = 75$?
 $75 \geq 80$? False
 $75 \geq 70$? True → B

```

if score >= 80:
    print('A')
elif score >= 70:
    print('B')
elif score >= 60:
    print('C')
else:
    print('D')

```

$\text{score} = 75$
 if score ≥ 80 : $75 \geq 80$ → False
 if score ≥ 70 : $75 \geq 70$ → True → B
 if score ≥ 60 : $75 \geq 60$ → True → C
 else:
 print('D')

if-elif-else

* sequential check. stop when the first condition is T

